UNIT 1 PRE-TEST

The Real Number System

Answer each question, then check your work.

Name _____ Date _____ Period _____

SCORE 3.0 Basic Understanding

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

1. Write as a decimal: $3/8$

= _____

e.g. 0.5

2. Write as a fraction: 0.25

= _____

simplest form, e.g. 1/4

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

3. Classify: $\sqrt{144}$

Classification: _____

type R or IR

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

4. Write as a fraction: $0.\overline{6}$ (the 6 repeats)

= _____

simplest form

5. Write as a decimal: $2\frac{5}{6}$ (2 and 5/6)

= _____

use ... for the repeating part

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

6. Classify: $\sqrt{20}$

Classification: _____

type R or IR

Simplify each expression first, then classify the result.

7. Classify: $\sqrt{3} \times \sqrt{27}$

Classification: _____

simplify, then R or IR

8. Classify: $\sqrt{32} \div \sqrt{2}$

Classification: _____

simplify, then R or IR

1.3 Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

9. Estimate: $\sqrt{40}$



10. Estimate: $\sqrt{95}$



SCORE 5.0 Advanced Content

1.2 Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

11. Classify: $-(\sqrt{49} - \sqrt{36})$

Classification: _____

simplify, then R or IR

12. Classify: $3 + \sqrt{5}$

Classification: _____

type R or IR

1.4 Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

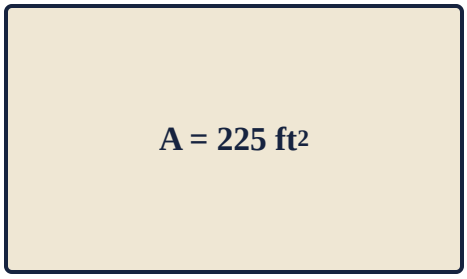
13. Set: { 0.6, $\sqrt{2}$, 1.5, 5/3 }



Order: _____

1.5 Area

A square garden has an area of 225 ft². Find the side length and the perimeter.



14. Find the side length and perimeter of the square garden.

Side length = _____ ft

Perimeter = _____ ft